

Micro-Expression Spotting with Face Alignment and Optical Flow

Wenfeng Qin

School of Computer and
Communication Engineering
University of Science and
Technology Beijing
Beijing, China
qinwenfeng666@163.com

Bochao Zou*

School of Computer and
Communication Engineering &
Shunde Graduate School
University of Science and
Technology Beijing
Beijing, China
zoubaochao@ustb.edu.cn

Xin Li

School of Computer and
Communication Engineering
University of Science and
Technology Beijing
Beijing, China
lixinustb424@gmail.com

Weiping Wang

School of Computer and Communication Engineering
University of Science and Technology Beijing
Beijing, China
weipingwangjt@ustb.edu.cn

Huimin Ma

School of Computer and Communication Engineering
University of Science and Technology Beijing
Beijing, China
mhmpub@ustb.edu.cn

ABSTRACT

Facial expression spotting holds significant importance as it can signify emotional changes. Particularly, micro-expressions possess the potential to reveal genuine emotions, making them even more valuable in practical domains such as public safety and finance. However, spotting micro-expressions proves challenging due to their subtle movements and brief duration. This paper proposes an expression spotting method based on face alignment and optical flow. We first use a finer crop-align technique to preprocess the facial videos by aligning the face and the nose tip. Then, regions of interest (ROIs) are defined by analyzing the statistics of action units. The optical flow features are then extracted and subjected to low-pass filtering to eliminate high-frequency noise. Furthermore, candidate expression segments are identified based on the magnitude of the processed optical flows. Finally, non-maximum suppression is utilized to remove overlapping segments. The effectiveness of the proposed method is evaluated on the challenge test set, resulting in an overall F1-score of 0.19. Additional results obtained from CAS(ME)² and SAMM Long videos provide further verification of the method's efficacy. The code is available online¹.

CCS CONCEPTS

• Computing methodologies → Computer vision tasks.

KEYWORDS: Expression spotting, Optical flow, Micro-expression, Macro-expression

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1 INTRODUCTION

Facial expressions can be categorized into macro-expressions (MaEs) and micro-expressions (MEs) based on their intensity and duration. MaE can be deliberately produced, whereas studies suggest that MEs can reveal genuine emotional reflections as they are brief, subtle, and unconscious expressions exhibited unintentionally by individuals [1]. Automatic ME spotting and recognition have wide potential applications in criminal interrogation, lie detection, political emotional analysis, and medical care [2]–[4], where accurate observation of genuine emotions is crucial. Locating the temporal boundaries of micro and macro expression segments through expression spotting serves as a prerequisite for subsequent recognition tasks.

Facial expression spotting can be categorized as a type of temporal action localization [5]. However, it differs from the latter in terms of lower amplitude and shorter duration of the "action". Moreover, the available datasets for expression spotting contain significantly fewer samples compared to those for temporal action localization. Consequently, data-driven deep learning methods have not achieved notable success in this task, as observed in previous challenges [6]. Conversely, traditional optical flow-based methods tend to perform better, especially when confronted with unseen datasets. Therefore, in our approach for the challenge, we also incorporate optical flow features.

Optical flow enables the capture of subtle movements between adjacent frames, which can serve as a representation of facial

¹ https://github.com/qin123xyz/MEGC2023_macro-and-micro-expression-spotting

* Corresponding author

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expression motion and aid in distinguishing between MaEs and MEs. To extract optical flow relevant to facial expressions, we initially perform face cropping and alignment to mitigate undesired variations in optical flow. Furthermore, to precisely capture optical flow corresponding to facial expressions, we define ROIs based on the Facial Action Coding System [7]. Specifically, the selection of ROIs is guided by the distribution of action units. Drawing inspiration from the elimination of overlap proposals in object detection, we employ an adapted non-maximum suppression (NMS) technique to remove and merge candidate expression segments.

2 RELATED WORK

The micro and macro-expression spotting task can be broadly classified into two categories: deep learning methods and traditional methods. Pan et al. [6] approached the spotting task by transforming it into a frame classification problem and utilized a convolutional neural network to classify videos into ME frames, MaE frames, and neutral frames. Wang et al. [8] proposed Spatiotemporal Convolutional Network and Clip Proposal Network to identify ME clips. Yang et al. [9] constructed a deep learning framework for the expression spotting task by leveraging facial action units. Leng et al. [10] employed optical flow as input for a convolutional neural network to classify frames. Despite the successful application of deep learning in various computer vision tasks, it heavily relies on a substantial amount of training data. The limited number of samples in micro-expression datasets poses a constraint on the performance of deep learning methods. Previous challenge results demonstrate that traditional methods outperform deep learning approaches when confronted with unseen test datasets. Notably, the winners of the MEGC 2021 and 2022 both employed traditional methods and did not rely on deep learning [11], [12], which spot facial expressions based on the spatial-temporal characteristics of ROI's optical flow.

3 METHODOLOGY

Expression spotting, particularly for MEs, is susceptible to interference from other movements like head movements and eye blinking. To accurately extract movement features relevant to MEs and MaEs, the proposed method consists of four distinct steps, as depicted in Figure 1: (1) Preprocessing: The original long video undergoes processing using sliding windows. Subsequently, the face region is cropped in each video frame, and face alignment is performed based on facial landmarks and nose tip area. (2) Feature extraction: ROIs are defined based on the statistical analysis of action units. Optical flows are then extracted within each ROI using the sliding window approach. (3) Expression segment generation: The extracted optical flow undergoes low-pass filtering to eliminate noise. The candidate apex frames and boundaries of expression segments are determined based on the magnitude of the optical flow. (4) Postprocessing: We merge certain candidate segments within sliding windows to obtain the final spotting results.

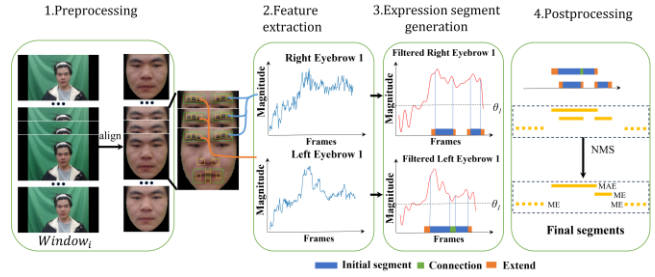


Figure 1: The overall pipeline of the proposed method.

3.1 Preprocessing

3.1.1 Face Cropping and Alignment. Face alignment plays a crucial role in preprocessing to eliminate interference from non-expression movements. Previous methods have addressed non-expression movement by subtracting the average motion across the image. However, this straightforward alignment technique only mitigates translation effects and cannot handle rotation and tilt. OpenFace [13] utilizing the Convolutional Experts Constrained Local Model for facial landmark detection and tracking, accurately detects facial landmarks and estimates head pose movement. Moreover, it iteratively projects 3D facial landmarks onto 2D facial landmarks to obtain the optimal rotation matrix and translation parameters, thus ensuring effective face alignment. Therefore, the OpenFace Toolkit is employed for face cropping and alignment.

3.1.2 Nose Tip Alignment. Since the aligned face obtained is processed for the entire long video, whereas optical flow is extracted within a sliding window that represents a subset of the video, we perform additional face alignment for each sliding window. Recognizing that the movement of the nose region can be seen as a global movement to some extent, we subtract the optical flow of the nose tip from each ROI.

3.2 Feature extraction

3.2.1 ROI Generation. To accurately capture the optical flow that corresponds to facial expressions, ROIs are defined based on the frequency of occurrence of action units in $CAS(ME)^2$ and $CAS(ME)^3$. Based on the analysis of the observed statistics of AUs, we have identified several high occurrence AUs that are relevant for defining ROIs. These include AU4, AU12, AU6, AU24, AU14, AU1, and AU2. The selected ROIs based on these AUs are visually represented with yellow rectangles in Figure 2. In addition, the several combinations of different ROIs are also considered as highlighted with green rectangles.

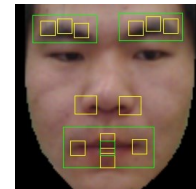


Figure 2: ROIs selection according to AUs statistics

3.2.2 Sliding Window. Considering that MaEs typically last from 0.5 seconds to 4 seconds, while MEs have durations shorter than 0.5 seconds, we employ sliding windows to divide the untrimmed long

videos into shorter segments. For facial videos with a frame rate of 30 frames per second (fps), We set the window size to 200 frames and compare two sliding strategies: fixed moving length with a 100-frame overlap and non-fixed moving length with an overlap ranging from 80 to 120 frames. In the non-fixed sliding strategy, the moving length is determined based on the onset and offset frame numbers of the candidate expression segment.

3.2.3 Optical Flow Extraction. We evaluate various optical flow estimation methods in terms of speed and accuracy. Figure 3 illustrates an example of the optical flow between the onset and apex frames. Among the methods compared, we select the Farneback [14] method to extract optical flow for each ROI within every sliding window. To quantify the magnitude of motion, we compute the norm2 of the optical flow vector for each pixel instead of using the vector directly. This allows us to obtain an optical flow curve, where the x-axis represents the frame and the y-axis represents the motion magnitude.

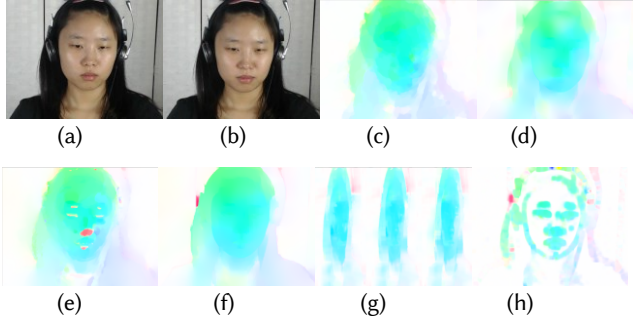


Figure 3: The raw picture and visualized optical flow using different methods (sub15_0502 in CAS(ME)²). (a) and (b) represent the onset and apex frames of an example expression, respectively. The optical flow methods depicted from (c) to (h) correspond to DeepFlow[15], PCAFlow[16], TVL1[17], SparseToDense[16], SimpleFlow[18], and Farneback[14], respectively.

3.3 Expression segment generation

3.3.1 Optical Flow Refinement. Despite employing landmark-based alignment and nose tip alignment, the obtained optical flow may still contain noise that is not solely caused by facial expressions. This noise can manifest as extremely small values or high-frequency variations. To address this, we compare several filtering methods, including low-pass filtering, convolve-average filtering, and Savitzky-Golay filtering. Figure 4 illustrates the comparison of these different filters for optical flow refinement. Notably, low-pass filtering proves effective in removing high-frequency noise while preserving the overall trend of the curve.

3.3.2 Candidate Segment Generation. After smoothing the optical flow curve, we define two threshold parameters for generating candidate expression segments. The first threshold, *threshold_1*, is utilized to exclude frames with optical flow values smaller than those expected for MEs. The remaining frames are grouped, and further filtering is performed to retain the frames where the difference between the optical flow intensity of the current frame and the minimum optical flow intensity within 30 frames exceeds

the *threshold_2*. Applying this two-step thresholding process yields candidate segments with the onset and offset frames.

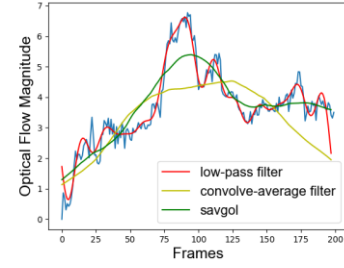


Figure 4: Comparison of different filters for optical flow refinement.

3.3.3 Apex and Boundary Localization. In the previous step, we ensure that the onset and offset frames of candidate segments are situated far from the valleys of the optical flow curve. In other words, the candidate segments obtained by *threshold_2* are short and centered around the peak frame. Therefore, we use the Apex frame to extend the boundaries and determine more suitable onset and offset frames. The pseudo-code for the boundary expansion algorithm is provided in Table 1.

Table 1. Pseudo Code of Boundary Expansion Algorithm

Input: $[on, off]$ represents the initial boundary of a candidate segment

1 Definition:

$$f_{max} = \max(F(on), \dots, F(off))$$

$$f_{onmin} = \min(F(\max(0, on - 30)), \dots, F(on), \dots, F(on + 30))$$

$$f_{offmin} = \min(F(\max(0, off - 30)), \dots, F(off), \dots, F(\min(len - 1, off + 30)))$$

$F(x)$ represents the optical flow value of the x -th frame, len is the length of the segment.

$$max = Index(f_{max})$$

$$onmin = Index(f_{onmin})$$

$$offmin = Index(f_{offmin})$$

where $Index(f)$ denotes the frame index of optical flow value, and max is the apex frame of $[on, off]$.

2. Expand the onset frame

if $onmin < on$:

for j in range $(on - 1, -1, -1)$:

if $F(j) - f_{onmin} < 0.33 * (f_{max} - f_{onmin})$:

$onset_{ex} = j$

break

if $F(j) > F(j + 1)$:

$onset_{ex} = j + 2$

break

else :

$$f_{onmin_{new}} = \min(F(\max(0, on - 10)), \dots, F(on + 1))$$

if $f_{on} - f_{onmin_{new}} > 0.3$:

$$Let\ onset_{ex} = Index(f_{onmin_{new}}) + 1$$

3. Expand the offset frame (similar to step 2) and obtain $offset_{ex}$

4: Output: Expanded segment $[onset_{ex}, offset_{ex}]$

3.4 Postprocessing

Taking inspiration from the proposal filtering process in object detection, we apply NMS to the candidate expression segments to merge or discard segments that are not expressions. The criteria of NMS in object detection are usually discarding entities that are below a given probability bound. Since our task lacks confidence scores for segments, we define Intersection over Union (IoU) as a replacement for the confidence score when performing NMS.

$$\text{IoU}(a, b) = \max\left(\frac{\text{len}(\text{Inter}(a, b))}{\text{len}(a)}, \frac{\text{len}(\text{Inter}(a, b))}{\text{len}(b)}\right) \quad (1)$$

where $\text{Inter}(a, b)$ denotes the intersection between segments a and b . $\text{len}()$ indicates the length of a segment.

4 EXPERIMENTS

4.1 Datasets and Evaluation Metric

We evaluate our method on CAS(ME)²[19], SAMM Long Video[20], and then test on SAMM-UNSEEN and CAS(ME)³-UNSEEN [21], [22] datasets. The fps of CAS(ME)² and CAS(ME)³ is 30, while the fps of SAMM is 200. For generalization purposes, we downsampled the SAMM dataset by selecting one frame from every 7 frames. The evaluation metric used in MEGC2023 is the F1-score, which is calculated based on the IoU of the predicted segment and the ground truth segment, as shown in equation (2). A predicted segment is regarded as true positive (TP) if $\text{IoU} > 0.5$, false positive (FP) otherwise. False negative (FN) refers to segments listed in the ground truth as expressions but not detected.

$$\text{IoU} = \frac{\text{Len}(\text{predict} \cap \text{ground truth})}{\text{Len}(\text{predict} \cup \text{ground truth})} \quad (2)$$

4.2 Hyperparameter Settings

Our method involves several hyperparameters, such as window size, moving size of the sliding window, threshold₁, threshold₂, and the NMS threshold. The range for each parameter is determined by analyzing its intended purpose. As mentioned in Section 3.3, threshold₁ aims to exclude optical flow values smaller than those expected for typical MEs. Hence, we set the range for threshold₁ from 0.9 to 2.0. Threshold₂ is intended to push frames within the candidate segment further away from the valleys of the optical flow curve and obtain shorter candidate segments concentrated around the Apex frame. The range for threshold₂ is set from 1.5 to 2.3. These threshold values are determined through experimentation. The NMS threshold in equation (1) aims to merge overlapping segments. Initially, the NMS threshold is set in the range of 0.2 to 0.5. Additionally, following [23], we also utilize the Bayesian algorithm hybridized with Nash equilibrium to search for the optimal parameters.

4.3 Spotting Results

We compare our method with other state-of-the-art (SOTA) spotting methods, including traditional and deep learning approaches. Tables 2 and 3 show the F1-scores of different spotting methods on the CAS(ME)² and SAMM long videos. Our proposed method demonstrates comparable performance to other SOTA methods, achieving the highest F1-score on CAS(ME)² and the best F1-score for ME spotting on the SAMM Long Video dataset. Furthermore, we evaluate our method on the test set of the MEGC 2023 challenge, fine-tuning the parameters through trial and error. The best results are presented in Table 4, where our method outperformed the baseline and secured the second place on the leaderboard.

Table 2: F1-scores of different spotting methods on CAS(ME)².

Method	Macro-expression	Micro-expression	Overall
He [11]	0.3782	0.1965	0.3436
MTSN [24]	0.4104	0.0808	0.3620
AUW-GCN [25]	0.4235	0.1538	0.3834
Ours	0.4445	0.1682	0.4066

Table 3: F1-scores of different spotting methods on SAMM.

Method	Macro-expression	Micro-expression	Overall
He [11]	0.4149	0.2162	0.3638
MTSN [24]	0.3459	0.0878	0.2867
AUW-GCN [25]	0.4293	0.1984	0.3728
ours	0.4168	0.2240	0.3680

Table 4: Results on the MEGC 2023 test set.

	F1-score (Overall)	F1-score (SAMM)	F1-score (CAS)
Baseline [26]	0.03	0.00	0.03
Ours	0.19	0.29	0.18

5 CONCLUSION

Considering the limited availability of samples for expression spotting and the successful track record of optical flow-based methods in recent challenge competitions, this paper presents a novel expression spotting method that leverages fine-level face alignment and optical flow refinement. The proposed method employs a more precise crop-align technique for preprocessing facial videos. ROIs are defined based on the analysis of action unit statistics. Optical flow features are extracted and subjected to low-pass filtering to eliminate high-frequency noise. Candidate expression segments are identified based on the magnitude of processed optical flows, and NMS is applied to determine the final expression segment. The effectiveness of the proposed method is evaluated on CAS(ME)² and SAMM Long videos, demonstrating promising results. Notably, the proposed method achieved the second-place ranking on the MEGC2023 test set.

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